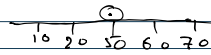


Measure of Central Tendency:-

Measure of central tendency are statistical metrics that describe the center point or a typical value of a dataset. They provide a single value that summarizes a set of data by identifying a central position within a dataset.



- i) Mean
- ii) Median
- iii) Mode

i) Mean

Def:- Sum of all observations divided by total no. of observations



Population mean (μ)

Sample mean ($\bar{x}$)

Population size (N)

Sample size (n)

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

x is a Random variable

$$x = \{5, 8, 12, 15, 20\}$$

$$N = 5$$

$$\mu = \frac{5 + 8 + 12 + 15 + 20}{5} = \frac{60}{5} = 12$$

Characteristics

- ⇒ Affected by outliers
- ⇒ used for Interval & Ratio

$$\{5, 8, 12, 15, 20, 100\}$$

ii) Median:-

The Median is the middle value of a dataset when values are arranged in ascending or descending

$$x = \{1, 2, 3, 4, 5\}$$

$$N = 5 \text{ (odd)}$$

$$\left(\frac{N+1}{2}\right)^{\text{th}} \text{ value} = 3^{\text{rd}} = 3$$

$$x = \{3, 4, 1, 5, 2, 100\}$$

$$= \{1, 2, 3, 4, 5, 100\}$$

$$N = 6 \Rightarrow 3^{\text{rd}} \text{ and } 4^{\text{th}}$$

$$= \frac{3 + 4}{2} = 3.5$$

Chars:-

- ⇒ Not affected by outliers
- ⇒ used for ordinal, Interval & Ratio

$$M, M, F, F$$

iii) Mode:-

The mode is a value that appears most frequent in a dataset

$$\text{Ex: } 2, 4, 4, 6, 7, 7, 7, 9$$

$$\text{Mode} = 7$$

$$\text{Ex: } 3, 5, 5, 6, 6, 8$$

$$\text{Mode} = (5, 6)$$

Bimodal

Chars:-

- ① Not affected by outliers
- ② useful for Nominal, ordinal, Interval and ratio data

Real world Application

Feature Selection Feature Engg

Age weight

salary

gender

degree

ordinal Rank

$$\text{Mean} \quad \left(\begin{array}{ccccc} 13 & 50 & 30k & M & B.E. 2 \end{array} \right)$$

	Age	weight	Salary	Gender	Degree	ordinal Rank
Men	18	50	30K	M	BE	2
	19	65	40K	F	BSC	3
	20	70	50	M	B Ed	2
	25	80	-	-	PHD	1
	-	-	-	-	-	-